

6

Civil Engineering Projects

UNIT SPECIFICS

Through this unit we have discussed the following aspects:

- **Civil Engineering projects and its impact**
 - Environmental Impact Analysis procedures
 - Sustainable Construction - Waste avoidance/ Efficiency increase
Advanced construction techniques for better sustainability
Techniques for reduction of Green House Gas emissions in various aspects of Civil Engineering Projects
- **Project Management and Contributions of Civil Engineers**
 - Paradigms & Systems - Waterfall / Traditional Project Management, Agile Construction Management, and Lean Construction Management
 - Quality of products, Health & Safety aspects for stakeholders
 - Demand and Contribution of Civil Engineers
- **Innovations and methodologies for ensuring Sustainability in Projects**

Besides giving a large number of multiple-choice questions as well as questions of short and long answer types marked in two categories following lower and higher order of Bloom's taxonomy, a list of references and suggested readings are given in the unit so that one can go through them for practice.

There is a "Know More" section, which has been carefully designed so that the supplementary information provided in this part becomes beneficial for the users of the book. It is important to note that for getting more information on various topics of interest some QR codes have been provided which can be scanned for relevant supportive knowledge. This section mainly highlights applications of the subject matter for our day-to-day real life or/and industrial applications on variety of aspects, case study related to environmental, sustainability, social and ethical issues whichever applicable, and finally inquisitiveness and curiosity topics of the unit.

RATIONALE

This concluding unit on civil engineering projects, offers the civil engineer a pragmatic view of the tasks and responsibilities to be shouldered during professional practice, be it with respect to environmental sustainability assessment and mitigation on site, or project management aspects.

UNIT OUTCOMES

List of outcomes of this unit is as follows:

U6-O1: Knowledge on Civil Engineering projects and its impact

U6-O2: Knowledge on Project Management and Contribution of Civil Engineering

U6-O3: Knowledge on Innovations and methodologies for ensuring Sustainability during Project development.

Unit-6 Outcomes	EXPECTED MAPPING WITH COURSE OUTCOMES						
	<i>(1- Weak Correlation; 2- Medium correlation; 3- Strong Correlation)</i>						
	CO-1	CO-2	CO-3	CO-4	CO-5	CO-6	CO-7
U6-O1	3	3	3	2	3	3	2
U6-O2	2	1	2	3	3	3	3
U6-O3	1	3	2	1	3	2	1

Civil Engineering projects are multi-faceted and to address its impact on the environment, various procedures, activities, and strategies are employed by large, multidisciplinary teams, sometimes across geographies. By embracing sustainable practices and technologies, civil engineering projects can minimize their ecological footprint and contribute to a more sustainable future.

6.1 CIVIL ENGINEERING PROJECTS AND ITS IMPACT

A construction project is a collaborative effort with “a group of interrelated work activities constrained by a specific scope, budget, and schedule to deliver capital assets needed to achieve the strategic goals of an Agency” (Shadan and Fleming, 2012), where the agency or client, maybe an individual, a private or public enterprise.

A construction project typically has the following phases (refer Fig. 1);

- **Project initiation** which entails, defining the project requirements, scoping the project, develop a project delivery strategy and contract management planning.
- **Planning/Environmental Clearance and Real Estate Acquisition** which entails, planning various site studies, checking compliance and applying for Environmental clearance, acquiring or relocating and taking possession of the land.
- **Design** entails designing the built environment as per stakeholder requirements and site conditions, design management, design reviews, feasibility/constructability reviews, risk assessment, Quality assurance and Quality control, incorporating Sustainability (Green building) standards and codes, and Certification.
- **Construction** entails construction management, third-party coordination, Quality management, Safety management during construction.
- **Commissioning** entails validating the building equipment and systems with operations personnel and third parties.
- **Project Close-out / Operations** is when the project begins to function as intended and all hand-over is done to the Agency/client.

The success of a project requires; Collaborating with architects, engineers and contractors to develop plans and establish timelines, Estimating and negotiating project costs, Creating and monitoring project budgets, Securing permits and design evaluations, Making schedules and work timetables, Determining what methods and strategies are appropriate for a project, Communicating with clients, contractors and other stakeholders, Assembling and leading construction teams, Working with building, construction and regulatory specialists, and Managing the day-to-day workflow of a project. The environmental impact of the project has to be systematically addressed and mitigated at each phase through procedure and a sustainable construction commitment.

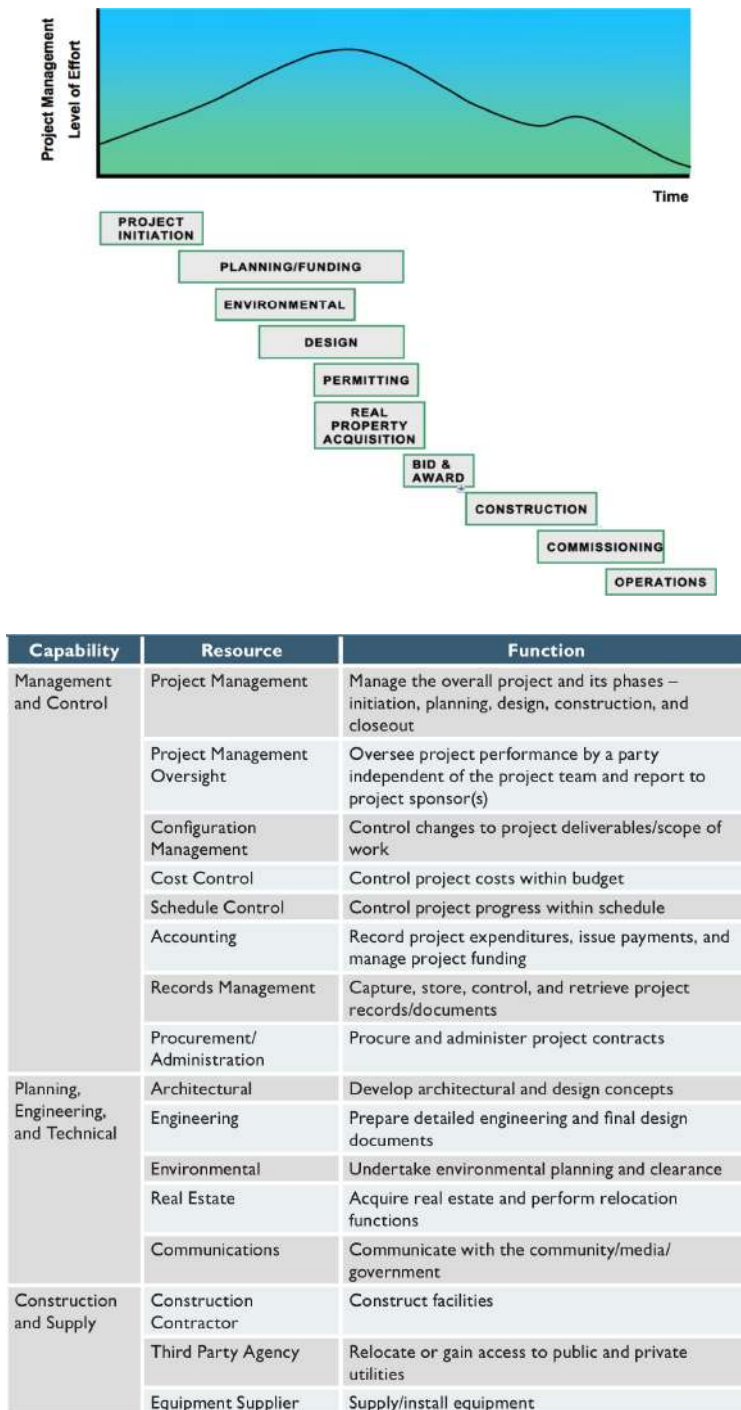


Fig. 6.1 : (Top) Typical Project Life Cycle - Traditional Design/Bid/Build, (Bottom) Project Resources (Shadan and Fleming, 2012)

6.1.1 EIA Procedures : Environmental Clearance for Project

During project planning and pre-design, a critical step is acquiring the **Environmental Clearance**.

The EIA Notification of 2006 mandated Prior Environmental Clearance, which involves four stages namely, screening; scoping; public consultation; and appraisal, for certain category of projects. The Ministry of Environment, Forest and Climate Change (MoEF), GoI, manages and publishes EIA Notifications and develops Sector specific Standard Terms of References (ToR) and manuals for the different sectors. The MoEF categorises all projects under ten sectors, i.e., (1) Mining, (2) Mineral Beneficiation, (3) Ports and Harbours, (4) Airports, (5-A) Building Construction, (5-B) Townships, (6) Asbestos, (7) Highways, (8) Coal Washery, (9) Aerial Ropeways, and (10) Nuclear power plants, Nuclear fuel processing plants and Nuclear waste management plants, and states that certain type (B) do not require to submit an EIA report for the clearance. EIA methodology, as explained in earlier Unit, has various steps, namely: Screening, Scoping, Impact Analysis, Mitigation, Reporting, Review of EIA, Decision-Making, and Post Monitoring, to develop a report on the state of affairs of the proposed project and its possible impact on the environment.



To begin an **Environmental Clearance for Project**, following are the steps outlined by the Centre for Science and Environment, India;

Step 1: Project proponent identifies the location of proposed plant after ensuring compliance with existing siting guidelines.

Step 2: The project proponent then assesses if the proposed activity/project falls under the purview of environmental clearance. If it is mentioned in schedule of the notification, the proponent conducts an EIA study either directly or through a consultant, however, B2 projects do not require preparation of EIA reports.

Step 3: After the EIA report is ready, the investor approaches the concerned State Pollution Control Board (SPCB) and the State Forest Department (if the location involves use of forestland). The SPCB evaluates and assesses the quantity and quality of effluents likely to be generated by the proposed unit as well as the efficacy of the control measures proposed by the investor to meet the prescribed standards. If the SPCB is satisfied that the proposed unit will meet all the prescribed effluent and emissions standards, it issues consent to establish (popularly known as NOC), which is valid for 15 years.

Step 4: The process of public hearing is conducted prior to the issue of NOC from SPCB. The District Collector is the chairperson of the public hearing committee. Other members of the committee include the official from the district development body, SPCB, Department of Environment and Forest, Taluka and Gram Panchayat representative, and senior citizen of the district, etc. The hearing committee hears the objections/suggestions from the public and after inserting certain clauses it is passed on to the next stage of approval (Ministry of Forest and Environment).

Step 5: The project proponent **submits an application for environmental clearance** with the MoEF if it falls under Project A category or the state government if it falls under project B category. The application form (1 and 1A, *details given below in next Section*) is submitted with EIA report, EMP, details of public hearing and NOC granted by the state regulators.

Step 6: Environmental appraisal: The documents submitted by an investor are first scrutinised by a multi-disciplinary staff functioning in the Ministry of Environment and Forests who may also undertake site-visits wherever required, interact with the investors and hold consultations with experts on specific issues as and when necessary. After this preliminary scrutiny, the proposals are placed before specially constituted committees of experts whose composition is specified in the EIA Notification.

Step 7: Issues of clearance or rejection letter: When a project requires both environmental clearance as well as approval under the Forest (Conservation) Act, 1980. The clearance granted shall be valid for a period of five years for commencements of the construction or operation of the project.

Application Forms for Environmental Clearance

Details of **Form1** is to be filled, which comprises of general project information and a checklist for confirmation on the Activity and Environmental Sensitivity of the proposed site is to be filled out, as below;

1. Basic Information, on

- 1.1. Name of project, Proposed capacity/area, Type of Project (New, Modernisation, Expansion), Existing capacity/area, Category of project, Location, etc.
- 1.2. Whether the proposal involves approval/clearance under: if, yes details of the same and their status to be given a) The Forest (Conservation) Act, 1980? (b) The wildlife (protection) Act, 1972? (c) The CRZ Notification, 1991?
- 1.3. Whether there is any Government Order/policy relevant/relating to the site?

- 1.4. Forest land involved (hectares)
- 1.5. Whether there is any litigation pending against the project and/or land in which the project is proposed to be set up?
2. **Activity** (*Each sub-section has several Information/Checklist confirmation*)
 - 2.1. Construction, operation or decommissioning of the Project involving actions, which will cause physical changes in the locality (topography, land use, changes in water bodies, etc.)
 - 2.2. Use of Natural resources for construction or operation of the Project (such as land, water, materials or energy, especially any resources which are non-renewable or in short supply):
 - 2.3. Use, storage, transport, handling or production of substances or materials, which could be harmful to human health or the environment or raise concerns about actual or perceived risks to human health.
 - 2.4. Production of solid wastes during construction or operation or decommissioning (MT/month)
 - 2.5. Release of pollutants or any hazardous, toxic or noxious substances to air (Kg/hr)
 - 2.6. Generation of Noise and Vibration, and Emissions of Light and Heat:
 - 2.7. Risks of contamination of land or water from releases of pollutants into the ground or into sewers, surface waters, groundwater, coastal waters or the sea:
 - 2.8. Risk of accidents during construction or operation of the Project, which could affect human health or the environment.
 - 2.9. Factors which should be considered (such as consequential development) which could lead to environmental effects or the potential for cumulative impacts with other existing or planned activities in the locality.
3. **Environmental Sensitivity**
 - 3.1. Areas protected under international conventions, national or local legislation for their ecological, landscape, cultural or other related value
 - 3.2. Areas which are important or sensitive for ecological reasons - Wetlands, watercourses or other water bodies, coastal zone, biospheres, mountains, forests
 - 3.3. Areas used by protected, important or sensitive flora or breeding, foraging, resting, over wintering, migration.
 - 3.4. Inland, coastal, marine or underground waters
 - 3.5. State, National boundaries
 - 3.6. Routes or facilities used by the public for access to recreation or another tourist, pilgrim areas
 - 3.7. Defence installations
 - 3.8. Densely populated or built-up area

- 3.9. Areas occupied by sensitive man-made land uses (hospitals, schools, places of worship, community facilities)
- 3.10. Areas containing important, high quality or scarce resources (ground water resources, surface resources, forestry, agriculture, fisheries, tourism, minerals)
- 3.11. Areas already subjected to pollution or environmental damage. (*Those where existing legal environmental standards are exceeded*)
- 3.12. Areas susceptible to natural hazard which could cause the project to present environmental problems (earthquakes, subsidence, landslides, erosion, flooding or extreme or adverse climatic conditions)

Then, Proposed **Terms of Reference** (TOR) for EIA studies applicable, is to be agreed by the proposer.

Details of **Form 1A**, which is a check list of environmental impacts is to be filled out, which is a table of 9 sections across the different environmental categories, which are;

1. Land Environment (1.1-1.9 questions)
2. Water environment (2.1-2.14 questions)
3. Vegetation (3.1- 3.3 questions)
4. Fauna (4.1- 4.3 questions)
5. Air environment (5.1- 5.6 questions)
6. Aesthetics (6.1- 6.4 questions)
7. Socio-economic aspects (7.1- 7.3 questions)
8. Building materials (8.1- 8.4 questions)
9. Energy conservation (9.1- 9.13 questions)
10. Environment management plan (for Construction phase and Functional /Use phase) across 5 subheadings; i.e., Environmental Components (Ambient Air, noise, water, Land Aesthetics), Predicted Impact, Probable source of Impact, Mitigation Measures and Remarks.



Fig. 6.1: Elements of a Sustainable Project

6.1.2 Sustainable Construction

Sustainable construction is defined as “*how the construction industry together with its product the ‘built environment’, among many sectors of the economy and human activity, can contribute to the sustainability of the earth including its human and non-human inhabitants*” (Kibert, 2007). The goal of sustainable construction is to minimize the environmental impact caused by the construction industry, through the following objectives:

1. Utilise renewable and recyclable materials.
2. Decrease the embodied energy within building materials.
3. Reduce the energy consumption of the completed building.
4. Minimize on-site waste generation.
5. Safeguard natural habitats.

Sustainable construction encompasses various activities, ranging from extracting materials and manufacturing products, to assembling them into buildings, maintaining and replacing systems, and ultimately disposing of waste, building systems, and the building structure, whilst considering the above mentioned objectives throughout the entire lifecycle of the construction process and the resulting built environment. Additionally, the physical distribution and relationships between buildings and infrastructure play a role in determining consumption patterns, which are influenced by planning decisions. The implementation of sustainable construction is further complicated by the involvement of public policy in the form of regulations, incentives, and disincentives. It also involves the participation of industries such as real estate, finance, and insurance, as well as institutions like higher education, design firms, and construction companies.

Waste Avoidance/ Efficiency Increase

Overall **efficiency increases**, in terms of material usage, manpower and equipment utilisation, energy and other resource consumption, and its associated economic costs, by waste avoidance. *Reducing disposal of waste* construction and demolition materials, reduces the environmental impact caused by the extraction and consumption of virgin resources and the production of new materials is offset. It brings about cost reductions in overall building projects by avoiding purchase costs through the reuse of materials and by donating recovered materials to qualified charities, which can provide tax benefits. Furthermore, the *use of onsite material reuse* helps lower transportation costs. Employing *deconstruction and selective demolition methods* leads to the creation of jobs, stimulates economic activities in recycling industries, and generates increased business opportunities within local communities. Last but not the least, *implementing sustainable construction practices* reduces the need for disposal facilities, thereby mitigating associated environmental issues, which in turn, contributes to the conservation of landfill space by diverting materials from disposal.

Techniques for reduction of Green House Gas emissions

The construction industry has several activities which result in direct or indirect Greenhouse gas emissions. The buildings and construction sector were responsible for 36% of total energy consumption and 39% of carbon dioxide (CO₂) emissions related to energy use and industrial processes. Out of these emissions, approximately 11% can be attributed to the manufacturing of building materials and products, including steel, cement, and glass, as per IEA (2018). Thus, the concept of embodied carbon captures the implied CO₂ equivalent/GHG emission, as discussed earlier, and is the critical decider when it comes to strategizing techniques to reduce GHG emission.

Techniques recommended by AIA are;

- ***Limit carbon-intensive materials***, like aluminium, plastics, and foam insulation, and use judiciously.
- ***Choose lower carbon alternatives***, such as, wood structure instead of steel and concrete, or wood siding instead of vinyl, and review Environmental Product Declarations for selecting alternatives.
- ***Choose carbon sequestering materials***, like wood, straw or hemp insulation and bring down the embodied carbon in a project.
- ***Reuse materials***, like brick, metals, broken concrete, or wood, as salvaged materials typically have a much lower embodied carbon footprint than those newly manufactured.
- ***Use high-recycled content materials***, particularly for metals as they are carbon-intensive but can be recycled, which brings down its embodied carbon value.
- ***Use fewer finish materials***, such as, polished concrete slabs in place of tiled, carpet or vinyl finished flooring saves the embodied carbon.
- ***Minimize waste***, as discussed above. Modularity and standard sizes for common materials like plywood, gypsum boards, wood framing, and pre-cut structural members can be factored in during design and wastage can be minimised.
- ***Maximize structural efficiency***, as it is the highest contributor to the embodied carbon of the project, and using optimum value engineering wood framing methods, efficient structural sections, and slabs are all effective methods to maximize efficiency and minimize material use.
- ***Reuse buildings instead of constructing new ones***, as renovation and reuse projects typically save between 50 and 75 percent of the embodied carbon emissions compared to constructing a new building.

Advanced Construction Techniques For Better Sustainability

Following are the technologies in construction have potential to significantly enhance sustainability;

1. **Prefabrication** involves the manufacturing of building elements/components or modules off-site and their assembly on-site, allowing for better quality control, reduced material waste, and faster construction timelines. This technique improves construction efficiency, reduces waste, enhances quality control, and minimizes disruption to the surrounding environment. It also enables easier deconstruction and material reuse at the end of a building's life cycle. In addition, employing a controlled environment in which to construct and pre-fabricate, results in improved quality of building components and less waste, as external conditions can no longer hinder its properties.



Fig. 6.2: Prefabricated building elements

2. **Modular construction** takes this a step further by assembling pre-made modules to create complete structures, offering flexibility, reduced on-site disruptions, and potential for deconstruction and reuse.
3. **Mass Timber Construction** refers to the use of large, prefabricated timber panels or components for structural elements, such as cross-laminated timber (CLT) and glued-laminated timber (glulam) which offer high strength, reduced carbon emissions, and faster construction times.
4. **Robotic construction** is an advanced technique involving robots and automation technologies, to perform various construction tasks, such as bricklaying, concrete pouring, and material handling, revolutionizing the construction industry. It offers high accuracy and efficiency, improved safety, enhanced productivity, and reduced labour costs.

5. **3D printing** utilizes robotic arms or gantries to precisely deposit layers of construction materials, such as concrete or specialized composite materials, based on complex and customised digital designs. 3D printing in construction offers several benefits, including faster construction timelines, reduced material waste, enhanced design flexibility, and cost savings, and additionally, allows for the integration of sustainable features, such as incorporating insulation directly into the printed structure.



Fig. 6.3 : Advanced technologies : (Top Left) Modular construction, (Top Right) Mass timber construction, (Bottom Left) Robotic construction and (Bottom Right) 3D printing

6.2 PROJECT MANAGEMENT

Project management in construction projects refers to the *discipline of planning, organizing, and controlling resources and activities to successfully deliver construction projects within defined constraints of time, budget, and quality*. It involves coordinating various stakeholders, managing risks, and ensuring the project progresses smoothly from initiation to completion. The PMC (project management consultant) includes the project manager, project & planning engineer, construction manager, site engineer, surveyor, Quality control engineer, Health and safety officer, etc. It is their responsibility to ensure timely and quality construction of projects.

6.2.1 Project Management paradigms and systems

Paradigms and Approaches

In construction project management, different **paradigms or approaches** are employed based on the project's unique characteristics and requirements, such as;

1. **Traditional/Waterfall Paradigm:** The traditional paradigm follows a linear sequence of project phases, where each phase is completed before moving on to the next. This includes distinct stages such as initiation, planning, design, construction, and handover. It emphasizes detailed planning, documentation, and a hierarchical management structure.
2. **Agile Paradigm:** Agile project management focuses on flexibility, adaptability, and iterative development. It emphasizes collaboration, frequent feedback, and continuous improvement. Agile methodologies, such as Scrum and Kanban, allow for dynamic project planning, quick decision-making, and shorter development cycles.
3. **Lean Construction Paradigm:** Lean construction aims to maximize value and minimize waste by streamlining processes and eliminating non-value-adding activities. It emphasizes continuous improvement, visual management, and the reduction of bottlenecks. Lean principles, derived from the manufacturing industry, are applied to improve efficiency and productivity in construction projects.

Other systemic approaches based on Project Delivery strategy are ;

1. **Design/Bid/Build** approach is when the owner manages the project, contracts out design to engineering consultants and construction to contractors, and retains a General Engineering Consultant (GEC), Construction Management (CM) and Program Management consultant (PMC).
2. **Design/Build** approach is when a single entity, typically a design-build contractor, is responsible for both the design and construction of the project. This integrated approach reduces coordination issues and promotes efficient communication, resulting in faster project delivery and potentially lower costs.
3. **Turnkey or D/B/O/M approach** is when a turnkey contractor is in charge from design conceptualisation to operation and maintenance.
4. **Integrated Project Delivery (IPD)** is a collaborative approach where all project stakeholders, including the owner, architects, engineers, contractors, and suppliers, work together from the project's inception. It emphasizes early involvement, shared decision-making, and risk/reward sharing, fostering collaboration, and improving project outcomes.
5. **Public-Private Partnership (PPP)** is a contractual arrangement between a public agency and a private sector entity. It combines public sector requirements with private sector innovation

and expertise. PPPs can help finance, develop, and operate infrastructure projects, leveraging private sector resources while sharing risks and rewards.

Project Management Systems

Cleland (1977) proposed a model for project management system, through the perspective of 'systems approach', with the project team at the focus interacting with the various functional sub-systems, as below;

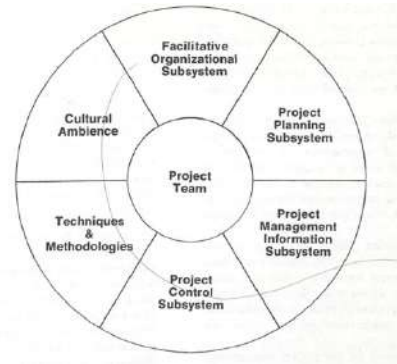


Fig. 6.4: Project management Model, showing the focal position of the project team and the interacting sub-systems (Cleland, 1977)

- The **Organizational Facilitative Subsystem** refers to the arrangement within an organization that combines project teams with the functional structure. This creates a "matrix" organization that establishes formal authority, responsibility patterns, and reporting relationships to facilitate the initiation and completion of specific projects. In this context, two key organizational units emerge: the project team and the functional units.
- The **Project Planning Subsystem** focuses on the selection of projects, identification of project objectives and goals, and the formulation of a strategy to achieve those objectives and goals. Project plans outline the necessary resources and allocation methods to support the project, drawing from the organization's resources regardless of their location.
- The **Project Control Subsystem** involves setting performance standards for the project's schedule, budget, and technical aspects. This subsystem incorporates feedback mechanisms to compare actual progress with planned progress and initiates corrective action when necessary. The control subsystem ensures effective monitoring of the various organizational units involved in the project, ensuring timely and within-budget project delivery.
- The **Project Management Information Subsystem** encompasses the essential intelligence required for effective project control. This subsystem can be informal, involving periodic meetings where project participants report on their project work's status. Alternatively, it can be a formal information retrieval system that provides regular updates on project activities. This subsystem enables project team members to make informed decisions and implement effective project management strategies.
- **Techniques and Methodology**, while not a subsystem in the traditional sense, encompass various management science techniques such as PERT, CPM, PERT-Cost related scheduling techniques, modeling, simulation, linear programming, and regression analysis. These techniques help evaluate risk and uncertainty factors in project decision-making.

- The ***Cultural Ambience Subsystem*** reflects the organization's environment and the practice of project management within it. The cultural ambience encompasses how individuals and social groups perceive and feel about project management practices in the organization. Factors such as emotions, attitudes, assumptions, experiences, and values shape the organization's cultural ambience. This ambience influences individual behaviours, thoughts, feelings, and expressions, ultimately determining socially acceptable behaviour within the organization.

Construction Project Management Software

The Global construction management software market is expected to expand at a CAGR (Compound Annual Growth Rate) of 8.70% from 2020 to 2027 and reach a net worth of \$2.73 billion by 2027, as per the (Data Bridge Market Research).

Construction project management software is a solution employed by professionals to streamline the construction planning process. These tools provide engineering estimates and capital maintenance capabilities, making them suitable for managing projects of varying sizes and complexities. It enables users to define categories or jobs, track itemized costs, and automatically generate financial reports. By incorporating features such as GIS/ESRI mapping, data warehouses, and budget and risk analysis, construction project management software facilitates smoother project development processes. Many construction ERP software packages include construction project management as a prominent feature.

Construction software aids in tracking project progress and monitoring offsite teams.

- ***Automated accounting*** functions streamline tasks such as accounts payables/receivables, payroll management, and work order entries.
- ***Job costing*** capabilities within construction management software enable builders and contractors to define costs associated with employees, production managers, contractors, and supervisors.
- ***Service management*** features in top-notch construction management software handle production schedules, work orders, and asset allocation.
- ***Scheduling*** for the project delivery lifecycle is critical, the tool helps with rescheduling timelines, managing resources, and updating schedules.
- ***Workload automation*** capabilities are beneficial for planning project timelines and balancing workloads across multiple channels, including predicting equipment and labor interdependencies.
- ***Reporting functionalities*** in construction project management software provide data summaries, evaluate project status, and present architectural services costs.

- **Inventory management** features in construction software help locate lost inventories, track assets on job sites, and conduct inventory audits.

There are several **construction project management software** presently available, each with certain pros and cons, briefly discussed below (<https://project-management.com/>) ;

Smartsheet provides a collaborative platform that enables users to work together, gather project requirements, and monitor initial cost estimates using a spreadsheet format. Teams can centralize project requirements and conveniently manage documentation by attaching files. The software also offers dashboards and portals to showcase key visual information relevant to the project. Additionally, users can utilize features such as project cost calculations, forms, and reports to enhance project management capabilities.

monday.com is a versatile platform designed to cater to the needs of architects, builders, general contractors, and engineers. It facilitates seamless collaboration and file sharing, enabling users to easily exchange files, images, updates, RFIs (Requests for Information), and feedback. The platform includes a built-in progress tracking feature that automatically monitors, calculates, and updates crucial information such as budgets, timelines, and resource allocation at every stage of the project. This data is presented through high-level dashboards and charts, providing users with a comprehensive overview of project performance. In addition, monday.com offers a range of valuable features including pre-built construction templates, tools for financial management, portfolio management capabilities, and efficient document management.

CoConstruct (Buildertrend) offers streamlined construction project management that incorporates a Schedule feature with Gantt charts and various views to ensure the smooth progress of all jobs. Users have the ability to create, assign, and monitor to-do tasks for team members and subcontractors, and can even set reminders for clients. The software allows for easy creation of to-dos through voice-to-text functionality, along with the ability to attach documents and photos and send automated notifications. Additionally, CoConstruct provides built-in communication tools for clients and trade partners, financial features for estimating and forecasting, capabilities for bidding and proposals, timesheet management, and seamless integration with other systems.

Fieldwire offers a communication platform that enables team members to engage in real-time conversations tied to specific tasks, promoting quicker decision-making and issue resolution. Users have the ability to monitor and manage all construction tasks, activities, and issues within the software. They can document and annotate issues, as well as update plans with detailed notes, photos, and videos. The construction project management software also provides scheduling and reporting features, along with mobile applications for on-the-go access. Field teams can effortlessly update plans and generate digital forms, streamlining the construction process.

Procore construction project management software incorporates mobile collaboration tools that foster task clarity, enabling teams to adhere to schedules and minimize rework. It captures and centralizes all project correspondence, ensuring easy accessibility. Real-time updates of information enhance visibility and mitigate risks. The software offers a comprehensive project overview, tracks all project steps, and expedites the approval process through features like a contact directory, document management, specification management, integrated scheduling, task management, Requests for Information (RFIs), drawing management, time card tracking, and email integration.

6.2.2 Quality of Products, Health and Safety aspects for stakeholder

Quality control and safety in construction projects are critical determinants of project success, and are intertwined with each other as a lack of quality may lead to a safety issue. Defects or failures in constructed facilities can lead to requirement for re-construction, thereby, affecting facility operations and causing project delays, cumulatively resulting in increased costs. It may also lead to accidents causing injury or fatality leading to further costs, such as incentive, insurance, inspection, etc. Specifying **quality requirements** in the design and contract documentation is crucial during the construction process. They should be well-defined and verifiable to ensure understanding and compliance among all project stakeholders. Decisions made during the planning and design phases have implications on both, quality and safety aspects as certain conformity and compliance can be built into the project management process from the start. However, both are subject to site conditions and other uncertainties, such as, change in the client request, or increase in material prices leading to the need of changing designs or revisiting decisions. Effective project managers strive to ensure that the work is done correctly from the beginning and that major accidents, at product and people level, are avoided. Beyond design decisions and planning vigilance, safety depends on education, alertness and cooperation amongst all stakeholders during the construction process.

Bugalia, Maemura and Ozawa (2019) note across their study comparing high-speed railways in India and Japan, that **safety is a culture** and that “*that top management must adopt a multi-pronged approach to improve the safety culture of an organization. There is no one-dimensional management strategy that is sufficient for improving the level of an organization’s safety culture*”. They further report that organisational reforms are required to bring about a shift in the safety culture and may be achieved through strong training systems, and recommend the need to study cultural aspects in tandem with technology, people and organisational aspects.



National and international bodies, such as, **National Occupational Safety and Health (OSH)** in India and Occupational Safety and Health Administration (OSHA) in the US, routinely conducts site visits of workplaces in conjunction with approved state inspection agencies. In India, Health and Safety is overlooked as per the '*The Building and other construction workers (Regulation of employment and conditions of service) Act 1996*', under which, Section 38 deals with Safety Committee and safety officers in every establishment; Section 39 deals with Notice of certain accidents (which causes death or any bodily injury by reason of which the person injured is prevented from working for a period of forty-eight hours or more) and prescriptions for the same; and Section 40 deals with Power of appropriate Government to make rules for the safety and health of building and other construction workers.

6.2.3 Demand and Contribution of Civil Engineering

According to the RICS (Royal Institution of Chartered Surveyors) Research on '*Real Estate and Construction Professionals in India by 2020*', industry projections, India's construction industry is predicted to reach a value of USD 1 trillion by 2030 and contribute around 13% to the country's GDP by 2025. It is the fastest growing sector and is on track to become the largest employer by 2022, providing jobs for over 75 million individuals. Presently, skilled (Supervisors, Technician, Foremen, Tradesmen, etc), semi-skilled and unskilled workers (helpers and Labourers) constitute 95.3% of the workforce, while only 1.2% is occupied by the Core Professionals, such as, Civil Engineers, Architects and Planners.

To fulfil the country's real estate needs (approximately 1.27 million civil engineers) and infrastructure requirements (approximately 3 million civil engineers), an annual average of 4 million civil engineers in the next decade is necessary, in spite of 1.5 million graduates in India every year. Due to the sustained shortage in annual supply and an increasing year-on-year demand, the cumulative demand for civil engineers between 2010 and 2020 is projected to be around 40.2 million. Unfortunately, there is an estimated shortfall of approximately 39.4 million civil engineers during the same period and recent surveys indicate a 6.27 percent decline in employment within the construction industry.

To Civil engineers play a **crucial role** in society by contributing to the development and maintenance of social infrastructure, as well as working towards building a sustainable future.

One of the key contributions of civil engineers is the planning, design, and construction of essential infrastructure such as roads, bridges, airports, water supply systems, and wastewater treatment plants. These infrastructure projects not only facilitate the movement of people and goods but also improve the overall quality of life by providing access to basic amenities and

services. Civil engineers ensure that these structures are safe, efficient, and environmentally sustainable, taking into account factors such as traffic flow, environmental impact, and resource efficiency.

Civil engineers also play a vital role in *addressing the challenges of urbanization* and population growth. They are involved in urban planning and the design of sustainable cities, taking into consideration factors like land use, transportation systems, and the efficient utilization of resources. By *integrating principles of sustainable development* into their projects, **civil engineers contribute to reducing environmental impact**, promoting energy efficiency, and creating resilient and liveable urban environments.

In the face of climate change, civil engineers are at the forefront of developing solutions to mitigate its effects and enhance resilience. They work on projects related to flood management, coastal protection, and the design of structures that can withstand natural disasters such as earthquakes and hurricanes. By *incorporating climate adaptation strategies* and utilizing innovative technologies, **civil engineers contribute to building a more resilient society** that can withstand the challenges posed by a changing climate.

Furthermore, **civil engineers contribute to the sustainable use of resources** through their focus on efficient design and construction practices. They promote the use of renewable materials, energy-efficient systems, and sustainable construction techniques, thereby minimizing the environmental impact of infrastructure projects. By embracing concepts such as *green building design and sustainable construction practices*, civil engineers help to reduce carbon emissions, conserve resources, and create environmentally friendly structures.

Moreover, **civil engineers actively engage in research and development** to drive innovation in their field. They explore *new materials, technologies, and methodologies* that can improve the efficiency, durability, and sustainability of infrastructure.

6.3 INNOVATIONS AND METHODOLOGIES FOR SUSTAINABILITY

While in the previous Units several technological and methodological innovations have been discussed, for project management a cutting-edge innovation is **Virtual Design and Construction (VDC)**. It is a technology that aids in the coordinated management of comprehensive performance models for design-construction projects, which encompasses various aspects such as the facilities themselves, work processes, and the organization of the design-construction-operation team. VDC involves using digital tools, such as Building Information Modeling (BIM), to visualize, simulate, and coordinate various aspects of a construction project. It enables better collaboration, clash detection, and data-driven decision-making, leading to improved project coordination, cost control, and reduced rework.

VDC offers several benefits, such as;

- **Integrated Approach with a Common Data Environment (CDE)** and allowing communication and collaboration with different project team members, third party members and other stakeholders.
- **Risk mitigation and Enhanced safety** for workers and end-users, as the level of detail allows decision-makers to identify potential hazards and mitigate them
- **Sustainability** that can be conceptualised and planned for, as VDC can support assessment of energy efficiency, carbon emissions, environmental impact, embodied carbon, etc. It can not only help incorporate strategies like, adaptive reuse of materials, but also aid design of improved IEQ and plan for future use.

The VDC is a combination of several tools and can to be tailored, such seen in the Singapore VDC Framework below which boasts the slogan “*Build Twice*” (*first Virtual, then real*) ;

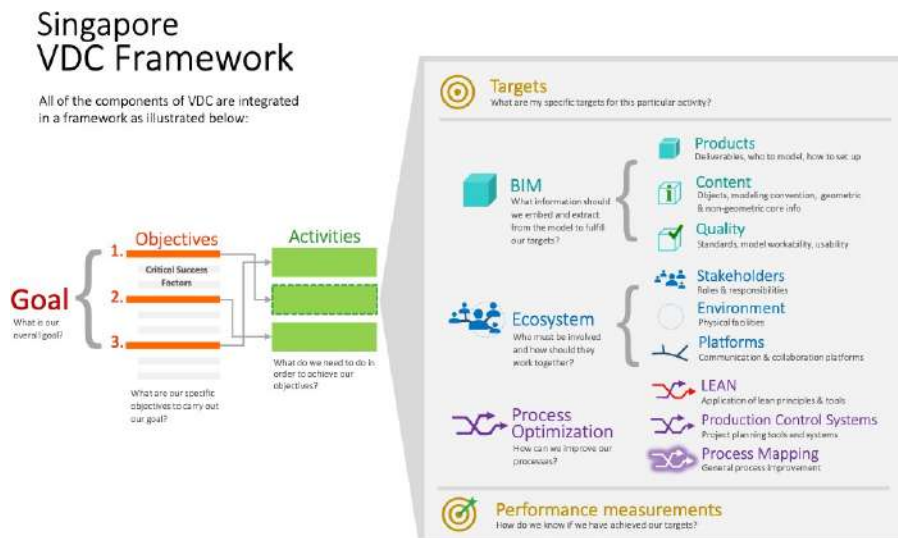


Fig. 6.5 : Singapore VDC Framework 2017 (www.corenet.gov.sg)

In essence, this technology – which is a culmination of all the understanding, knowledge, discussions, and practical applications so far explored in this book, bringing sustainability hand in hand with design and construction, is the apt point for reflecting on the global and societal impact of civil engineering and the great onus on the future civil engineer.

UNIT SUMMARY

This concluding unit on projects in civil engineering covers the procedures and techniques directly applicable to construction and related activities during practice. The specific procedure for ‘environmental clearance’ that is mandated in India as per EIA Notification, 2006, is outlined and Sustainable construction practices, with techniques for reducing GHG emissions and improving sustainability are covered to inform the requirements and consideration while proposing a project. Further, the elements of Project management, its paradigms, responsibilities, and overall contributions are discussed to give an overview on the systemic connections between decision-making during a project. Finally, the latest innovations for supporting projects are presented to make the aspiring civil engineer aware of the current trends in the domain.

EXERCISES

- Q. 6.1 What is an important element of an Environment Management Plan?
- (a) Probable source of Impact & Predicted Impact
 - (b) Environmental components (ambient air, noise, water, land aesthetics)
 - (c) Mitigation Methods
 - (d) all of the above
- Q. 6.2 Which of the following are not objectives of sustainable construction?
- (a) Utilise renewable and recyclable materials
 - (b) Green Building Certification
 - (c) Minimize on-site waste generation
 - (d) Safeguard natural habitats
- Q. 6.3 Which of the following techniques for reduction of GHG emissions requires Environmental Product Declaration information?
- (a) Minimising waste
 - (b) Limiting carbon-intensive materials,
 - (c) Choosing lower carbon alternatives
 - (d) Reusing materials

Q. 6.4 What is the traditional paradigm for Project management, where a linear sequence of phases is followed, called?

- (a) Agile paradigm
- (b) Lean construction paradigm
- (c) Design-Build paradigm
- (d) Waterfall paradigm

Q. 6.5 Which Indian National body instituted 'The Building and other construction workers (Regulation of employment and conditions of service) Act, 1996'?

- (a) National Occupational Safety and Health (OSH)
- (b) The Ministry of Environment, Forest and Climate Change (MoEF)
- (c) Occupational Safety and Health Administration (OSHA)
- (d) Environmental Protection Agency (EPA)

Answers of Multiple Choice Questions: 6.1 (d) , 6.2 (b) , 6.3 (c), 6.4 (d), 6.5 (a)

II. Short and Long Answer Type Questions

Q. 6.6 Why is Environmental impact Assessment and Environmental Clearance important consideration in the construction of built-environment.

Q. 6.7 What is Project management? Illustrate the different types of construction management paradigms and opine which is most suitable.

Q. 6.8 Discuss the various steps involved to get Environmental Clearance for a project in India. When is preparation of an EIA report not needed?

Q. 6.9 What is Sustainable Construction? Elaborate on the various strategies and techniques to reduce GH emissions and improve overall sustainability.

Q. 6.10 What are some of the new and innovative innovations and methodologies for sustainability incorporation into construction projects? Explain anyone that can be considered as a way of the future to ensure environmental sustainability.